

# FragmentIQ: The Systematic Bias of Image-Based Muckpile Fragmentation Analysis, and Its Correction

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CAOS open-research programme · CAOS\_FragmentIQ, image-based muckpile fragmentation analysis

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Not peer reviewed. Source code, data and computational records: [github.com/fsantibanezleal/CAOS\\_FragmentIQ](https://github.com/fsantibanezleal/CAOS_FragmentIQ)

**Abstract.** The size distribution of a blasted muckpile sets the energy cost of every downstream crushing and grinding stage, and it is routinely estimated by photographing the muckpile and analysing the image. The biases of this image-based granulometry are known qualitatively but rarely measured, because the true size distribution of a real muckpile is unknown. FragmentIQ measures them on synthetic muckpiles whose true fragmentation is known by construction, using the threshold and marker-controlled watershed delineation of the standard tools. Over seven cases the watershed finds 1.5 to 2.5 times as many fragments as exist, and the recovered  $P_{50}$  and  $P_{80}$  fall below the truth in every case, by 16.5–31.5% and 13.5–27.6%. About half of the shortfall is present before delineation: a perfect delineation of the ground-truth label map already recovers  $P_{50}$  7.9–15.8% below the nominal size, because the visible area of each fragment is smaller than its nominal diameter implies. The shortfall is systematic. A learned multiplicative correction, predicted from four shape features of the recovered distribution, reduces the mean  $P_{50}$  error on 17 held-out muckpiles from 28.4% to 3.98%. Refining the foreground with a convolutional boundary classifier before the same watershed lowers the mean  $P_{50}$  error on eight held-out muckpiles only from 27.2% to 23.8%, against 13.3% for a perfect delineation. A raw image-based fragmentation estimate is therefore biased fine by tens of percent; a size calibration against a known reference removes most of that bias, and a better delineation cannot remove the part present before delineation.

**Index Terms.** Rock fragmentation, muckpile, particle size distribution, image analysis, watershed segmentation, Rosin-Rammler, Kuz-Ram, blasting, calibration, synthetic ground truth.

## I. INTRODUCTION

THE degree to which a drill-and-blast round fragments the rock, quantified as the particle-size distribution of the resulting muckpile, is one of the most consequential numbers in a mine: a finer muckpile is cheaper to dig, haul and, above all, crush and grind, so blast design is tuned to it, and predicting it from blast parameters is the classic Kuz-Ram problem [1], [2]. Measuring it after the fact, by sieving, is impractical at production scale, so the standard practice is

image-based granulometry: photograph the muckpile, delineate the visible fragments, convert their areas to sizes, and fit a Rosin-Rammler distribution [3]. Commercial systems such as WipFrag [5] and Split delineate the fragments by edge detection and watershed segmentation [4]. Their error modes, the splitting of large blocks into several apparent fragments and the loss of fines below the image resolution, are usually stated as caveats rather than measured, because the true distribution of a real muckpile, against which the error would be measured, is not known.

FragmentIQ measures the error on synthetic muckpiles whose true fragmentation is known by construction. It runs the threshold and watershed delineation of the classical tools, reads  $P_{50}$  and  $P_{80}$  off the recovered size distribution, and compares them with the truth over seven cases that span coarse to fine and monosized to broad distributions. It then separates the part of the shortfall that is present before delineation from the part the delineation adds, measures how much of the shortfall a learned size correction removes, and tests whether refining the foreground with a learned boundary classifier closes the gap. The contribution is a ground-truthed measurement of the bias of the standard method, not a new fragmentation algorithm.

## II. SYNTHETIC MUCKPILES AND THE MEASUREMENT

**Scenes.** Each synthetic muckpile is a  $560 \times 420$ -pixel image of irregular polygonal fragments on a dark background. Fragment diameters are drawn from a Rosin-Rammler distribution  $P(x) = 1 - \exp[-(x/x_c)^n]$  with characteristic size  $x_c$  and uniformity index  $n$ . Each fragment is drawn as a polygon of 8 to 11 vertices at 0.78 to 1.12 times its nominal radius, with its own grey level, radial shading and grain, and the fragments are placed largest first so that they touch with little overlap, until the surface is full. The ground truth is the generator's own record: the nominal diameter of every placed fragment and a per-pixel map of the fragment each pixel belongs to. The seven cases (Table I) span three size regimes, the same medium muckpile under even and under raking light (I-EVEN, I-SHADOW), a monosized control whose fragments all have a nominal diameter of 120 mm (C-MONO), and a known Rosin-Rammler target with  $x_c = 160$  mm and  $n = 1.7$  (C-KNOWN).

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Interactive tool and reproducible pipeline (MIT) at [https://github.com/fsantibanezleal/CAOS\\_FragmentIQ](https://github.com/fsantibanezleal/CAOS_FragmentIQ); the tool is available at <https://fragmentiq.fasl-work.com>.

*Classical pipeline.* The foreground is the set of pixels brighter than a fixed grey threshold (58 of 255). A chamfer distance transform of the foreground gives one marker per local maximum after non-maximum suppression, and a marker-controlled flood in descending distance order labels the fragments and leaves ridges at their boundaries, the watershed of [4] as the classical tools apply it. Each labelled region gives an equivalent-circle diameter; the sizes form a mass-weighted cumulative passing curve, with mass proportional to the cube of the diameter, and  $P_{50}$  and  $P_{80}$  are read off that curve by linear interpolation. A Rosin-Rammler fit is made to the same curve. The true distribution is reduced in the same way from the nominal diameters.

*Perfect delineation.* To separate the error of the delineation from what the image can show at all, the same reduction is applied to the ground-truth label map: the equivalent-circle diameter of the pixels each fragment occupies in the image. This is the distribution a delineation would recover if it found every visible fragment exactly.

*Learned size correction.* A small regression network, with one hidden layer of 16 units, predicts a multiplicative correction  $k = P_{50}^{\text{true}}/P_{50}^{\text{rec}}$  from four features of the classical recovery: the recovered  $P_{50}$ , the ratio  $P_{80}/P_{50}$ , the logarithm of the fragment count, and the passing fraction at half the recovered  $P_{50}$ . It is trained on 39 of 56 synthetic muckpiles (four size regimes, two lighting conditions and seven seeds, all disjoint from the seven cases) and evaluated on the remaining 17.

*CNN-refined foreground.* Learned delineation of rock fragments has been demonstrated with deep instance segmentation [6]. Here a two-layer convolutional network classifies  $16 \times 16$ -pixel patches as inter-fragment seam or fragment interior, and it is used only to refine the foreground that the same watershed floods. It is trained on patches from the same 56 muckpiles, seam-centred patches against deep-interior patches with the ambiguous band between them excluded, and reaches a patch-level boundary F1 of 0.997 on a held-out fifth of the patches. The refined foreground is the thresholded foreground after a morphological closing, cut wherever the network's seam probability reaches a threshold. The grey threshold and the seam threshold (61 and 0.7) are selected on eight tuning muckpiles, and the result is measured on eight test muckpiles with disjoint seeds (four size regimes, two seeds each, even lighting).

### III. THE SYSTEMATIC BIAS

On every one of the seven synthetic muckpiles the recovered  $P_{50}$  and  $P_{80}$  fall below the true values (Fig. 1a, Table I): by 16.5 to 31.5% at  $P_{50}$  (median 23.3%) and by 13.5 to 27.6% at  $P_{80}$  (median 25.5%). The largest absolute shortfall is at the coarse end, where the true  $P_{80}$  of 1096 mm in R-COARSE is recovered as 794 mm. Raking light increases the  $P_{50}$  shortfall of the same muckpile from 20.9% (I-EVEN) to 23.6% (I-SHADOW), and even the monosized control C-MONO, whose true  $P_{50}$  and  $P_{80}$  are both 120 mm, is recovered at 97 and 104 mm. The watershed *over-segments* (Fig. 1b): it finds 1.5 to 2.5 times as many fragments as exist

TABLE I  
RECOVERED VERSUS TRUE FRAGMENTATION OVER THE SEVEN SYNTHETIC MUCKPILES ( $x_c$ ,  $n$ : THE ROSIN-RAMMLER PARAMETERS OF THE GENERATOR; T/R: TRUE/RECOVERED). THE WATERSHED FINDS MORE FRAGMENTS THAN EXIST AND RECOVERS  $P_{50}$  AND  $P_{80}$  BELOW TRUTH IN EVERY CASE. I-SHADOW IS THE I-EVEN MUCKPILE UNDER RAKING LIGHT.

| Case     | $x_c$ (mm), $n$ | found/true | $P_{50}$ t/r (mm) | $P_{80}$ t/r (mm) |
|----------|-----------------|------------|-------------------|-------------------|
| R-COARSE | 320, 1.4        | 116/70     | 836/698           | 1096/794          |
| R-MEDIUM | 180, 1.6        | 145/98     | 356/260           | 416/310           |
| R-FINE   | 90, 1.9         | 211/94     | 164/126           | 188/144           |
| I-EVEN   | 180, 1.6        | 163/81     | 310/245           | 374/274           |
| I-SHADOW | 180, 1.6        | 157/81     | 310/237           | 374/271           |
| C-MONO   | 120, mono       | 221/107    | 120/97            | 120/104           |
| C-KNOWN  | 160, 1.7        | 173/68     | 397/272           | 611/458           |

TABLE II  
WHERE THE SHORTFALL ARISES, IN PERCENT: THE SHORTFALL OF A PERFECT DELINEATION OF THE GROUND-TRUTH LABEL MAP AGAINST THE NOMINAL TRUTH (BEFORE DELINEATION), AND THE FURTHER SHORTFALL OF THE WATERSHED AGAINST THAT PERFECT DELINEATION (ADDED BY DELINEATION). THE TWO COMPOSE MULTIPLICATIVELY INTO THE SHORTFALLS OF TABLE I.

| Case     | $P_{50}$ shortfall (%) |       | $P_{80}$ shortfall (%) |       |
|----------|------------------------|-------|------------------------|-------|
|          | before                 | added | before                 | added |
| R-COARSE | 7.9                    | 9.3   | 10.5                   | 19.0  |
| R-MEDIUM | 12.0                   | 17.1  | 12.0                   | 15.3  |
| R-FINE   | 14.6                   | 10.2  | 11.2                   | 13.9  |
| I-EVEN   | 13.1                   | 8.9   | 14.2                   | 14.7  |
| I-SHADOW | 13.1                   | 12.1  | 14.2                   | 15.6  |
| C-MONO   | 10.9                   | 9.5   | 7.7                    | 6.3   |
| C-KNOWN  | 15.8                   | 18.6  | 14.0                   | 12.8  |

(median 2.0), because the grain and shading inside a fragment leave dark holes in the thresholded foreground, and each hole adds distance-transform maxima that seed extra regions. More, smaller apparent fragments shift the size distribution fine.

Over-segmentation is not the whole shortfall (Table II). A perfect delineation of the ground-truth label map already recovers  $P_{50}$  7.9 to 15.8% below the nominal truth (median 13.1%) and  $P_{80}$  7.7 to 14.2% below it (median 12.0%). Two properties of the scenes produce this part. Each fragment is drawn as a polygon within 0.78 to 1.12 times its nominal radius, so its area is on average smaller than that of the nominal circle, and fragments placed later overlap the edges of the earlier, larger ones, whose visible area shrinks further. The delineation then adds 8.9 to 18.6% at  $P_{50}$  (median 10.2%) and 6.3 to 19.0% at  $P_{80}$  (median 14.7%) relative to the perfect delineation. A raw image-based  $P_{50}$  is therefore an underestimate of tens of percent, with one part that a better delineation can remove and one part that it cannot.

### IV. CORRECTING THE BIAS

Because the shortfall is systematic, it is largely correctable (Fig. 2a). On the 17 held-out muckpiles the learned size correction reduces the mean absolute  $P_{50}$  error from 28.4% for the classical pipeline to 3.98%, a sevenfold reduction: the fine shift is consistent enough that one multiplicative factor, adapted to the shape of the recovered distribution, largely undoes it. A better delineation helps less (Fig. 2b). On

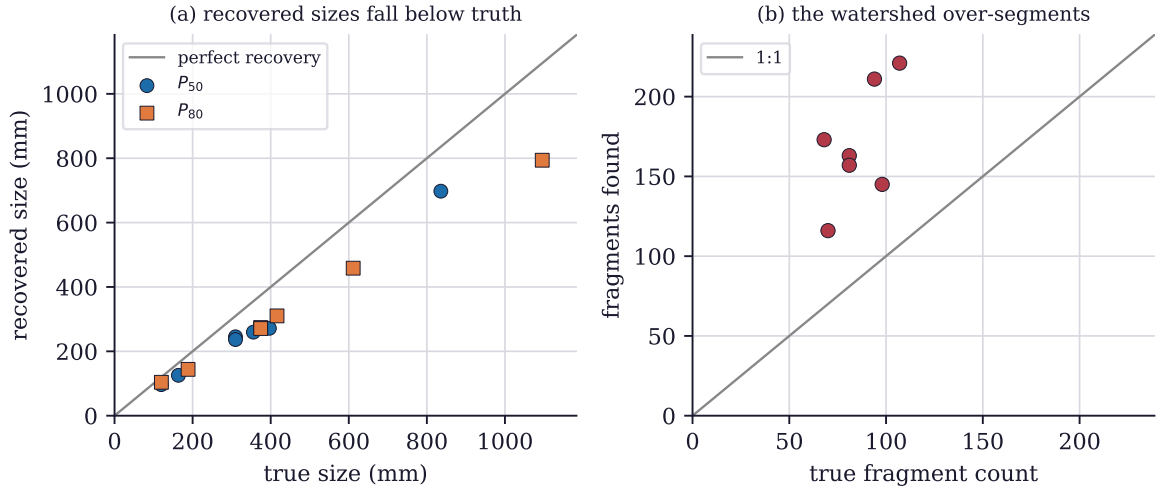


Fig. 1. The bias of image-based fragmentation analysis, measured against known ground truth over seven synthetic muckpiles. **(a)** Recovered versus true  $P_{50}$  and  $P_{80}$ : every point falls below the line of perfect recovery, so the pipeline reports sizes finer than the truth, by 16.5–31.5% at  $P_{50}$  and 13.5–27.6% at  $P_{80}$ . **(b)** Fragments found versus true fragment count: every point lies above the one-to-one line. The watershed *over-segments*, finding 1.5 to 2.5 times as many fragments as exist, which shifts the size distribution fine.

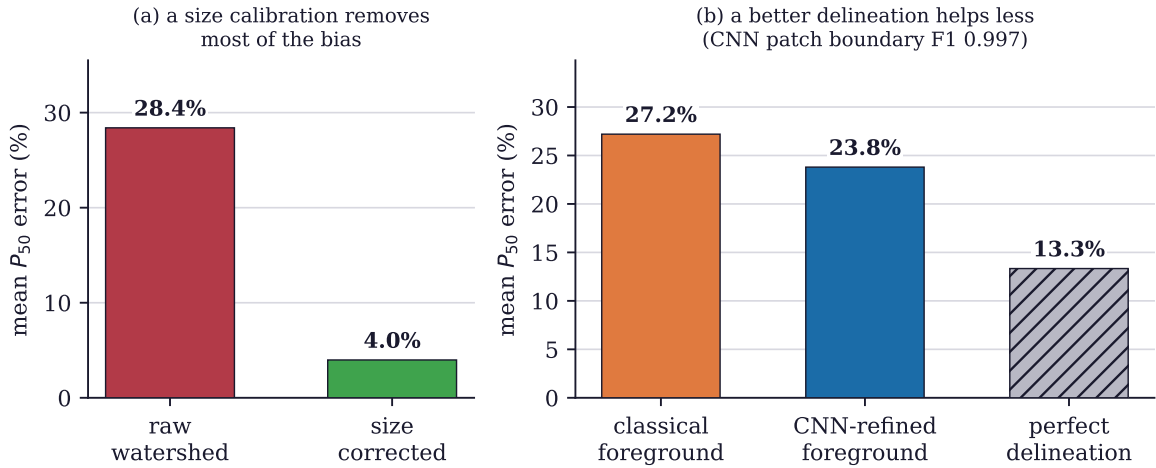


Fig. 2. Correcting the bias. **(a)** Mean absolute  $P_{50}$  error on 17 held-out muckpiles, for the classical pipeline and after the learned size correction: from 28.4% to 3.98%. **(b)** Mean absolute  $P_{50}$  error on eight held-out test muckpiles when the same watershed floods the classical foreground (27.2%), the CNN-refined foreground (23.8%), or a perfect delineation of the ground-truth label map (13.3%). The shortfall that a perfect delineation leaves is present before delineation, so only the size correction removes it.

the eight test muckpiles the CNN-refined foreground lowers the mean  $P_{50}$  error of the same watershed from 27.2% to 23.8%. A perfect delineation of the label map would still leave 13.3% on these muckpiles, because a correct segmentation cannot restore fragment area that the image does not show; the refined foreground closes 3.4 of the 13.9 percentage points between the classical pipeline and that limit. The size correction removes both parts at once, because it is fitted against the true size directly. In practice, the size estimate should first be calibrated against a known reference, and a better segmentation is a secondary improvement.

## V. DISCUSSION AND SCOPE

Image-based fragmentation analysis by threshold and watershed delineation is biased fine, here by 16.5 to 31.5%

at  $P_{50}$  and 13.5 to 27.6% at  $P_{80}$ . Roughly half of the  $P_{50}$  shortfall is present in the image before any delineation, and over-segmentation adds the rest. A learned size correction removes most of it, taking the mean  $P_{50}$  error to about four percent, while a learned refinement of the foreground helps only modestly, and even a perfect delineation leaves the first part. A raw muckpile image number is therefore an underestimate to be calibrated, and the calibration, not the segmentation method, is where most of the accuracy is won.

The measurements have limits. All scenes are synthetic, used because their true size distribution is known and the bias can therefore be measured, which is not possible on a real muckpile without independent sieving. The split of the shortfall between the image and the delineation depends on how the generator draws fragments and on taking the nominal diameter

as the true size; on a photograph the reference is a sieve size, whose relation to the visible outline differs. The held-out muckpiles of the size correction share the generator, size regimes and lighting of its training muckpiles and differ only in seed, so 3.98% is an in-distribution figure, and the patch F1 of the boundary classifier is measured on patches drawn from its training scenes. The magnitudes are representative of the mechanism, not a calibration transferable to a specific site, which would need its own ground-truthed reference.

## VI. CONCLUSION

Measured against ground truth on synthetic muckpiles, image-based fragmentation analysis by threshold and watershed delineation is biased fine: it finds 1.5 to 2.5 times the true fragment count and recovers  $P_{50}$  and  $P_{80}$  13.5 to 31.5% below truth. About half of the  $P_{50}$  shortfall is present before delineation, in the visible area of the fragments, and over-segmentation adds the rest. The bias is systematic, so a learned size correction reduces the mean  $P_{50}$  error from 28.4% to 3.98% on held-out muckpiles, while a learned refinement of the foreground reduces it only from 27.2% to 23.8%, against 13.3% for a perfect delineation. Calibrating the size estimate against a known reference, not improving the segmentation, is the dominant correction.

## APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry the seven cases' true and recovered size distributions and fragment counts (`case-results.json`) and the learned-model metrics (`fq-learned.json`: the raw and

size-corrected  $P_{50}$  errors, the classical and CNN-refined errors, and the boundary F1). The training, tuning and test muckpiles are generated by `data-pipeline/pipeline/science/gen_train.mjs`; the two networks are trained by `train_frag.py`, and the refined foreground is scored by `eval_frag.mjs`, both in the same folder. The script `manuscripts/fragmentation/data/decompose.mjs` runs the same engine on the seven cases and the eight test muckpiles and writes the shortfall decomposition (`decomposition.json`). The figure script under `manuscripts/fragmentation/figures/` reads `fq.json` and `decomposition.json` from that data folder. The generator, the watershed and the size reductions are deterministic given a seed, so re-running reproduces Tables I and II and Figs. 1 and 2.

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